1. Perceptron learning algo (supervised learning algo):

A. supervised learning: when training a model (neuron network), we will offer not only "inputs x_i" but also corresponding "output y (ground truth)" as pair training example [inputs x_i, output y (ground truth)] for the model to learn.

B. processes of algorithm: step1. randomly initialize the weights w_i.

step2. select one pair training example [inputs x_i] for perceptron to compute corresponding "predicted output P_y"

step3. predicted output P_y ? output y (ground truth):

```
# if "( p_y <0 ) != ( y = +1 ) " = " ( p_y = -1 ) != ( y = +1 ) "
 ,then " add each w[i] by η ∗ x[i]" = " w[i] += ( y ∗ η ∗ x[i] ), y = +1"

# if " ( p_y >0 ) != ( y = -1 )" = " ( p_y = +1 ) != ( y = -1 ) "
 ,then " subtract each w[i] by η ∗ x[i]" = " w[i] += ( y ∗ η ∗ x[i] ), y = -1"
```

step4. repeat "step2 & step3" until predicted output P_y = output y (ground truth).

2. Learning rate (η): an example of hyperparameter, which is not a parameter adjusted by learning algo but by user.

3. Decision boundary equation:

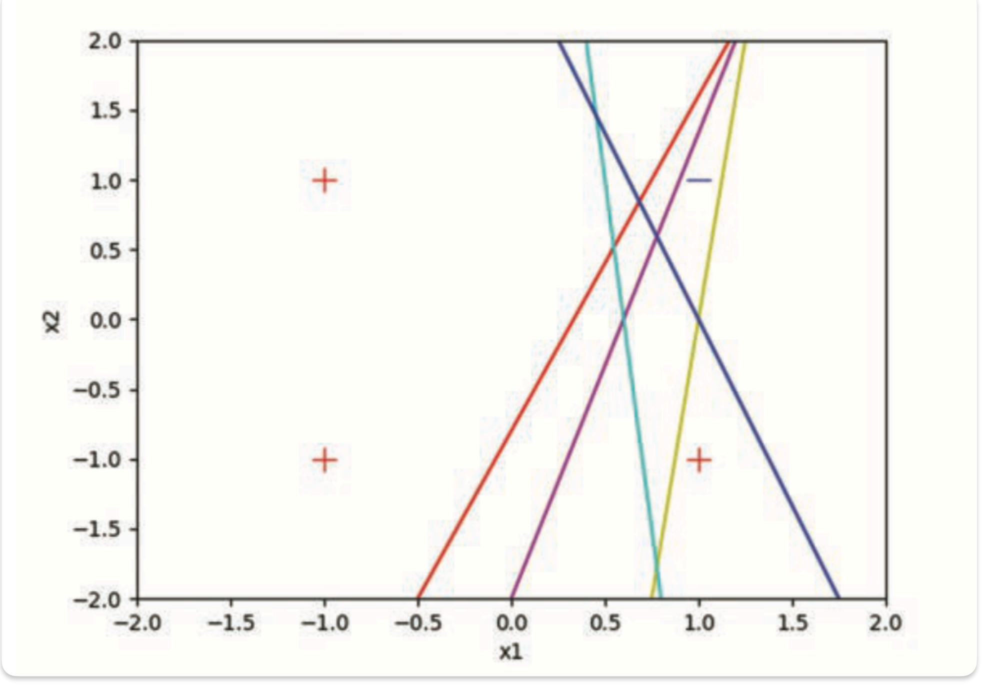
A. In two-input perceptron, there is a line ($w_0 x_0 + w_1 x_1 + w_2 x_2 = 0$) in 2D graph, the line can divide the graph into 3 parts $\Longrightarrow$ Part1. on the line: $w_0 x_0 + w_1 x_1 + w_2 x_2 = 0$

Part2. one side of the line: $w_0 x_0 + w_1 x_1 + w_2 x_2 > 0$

Part3. the other side of the line: $w_0 x_0 + w_1 x_1 + w_2 x_2 < 0$

B. the line: $w_0 \cancel{x_0}^{1} + w_1 x_1 + w_2 x_2 = 0 \longrightarrow x_2 = \underbrace{\boxed{-\frac{w_1}{w_2}}}_{\text{slope of line}} x_1 \underbrace{\boxed{-\frac{w_0}{w_2}}}_{y\text{-intercept of line}}$

C.



$\Longrightarrow$ the red line (built by initial weights) cannot divide the graph (one side for output=⊕, the other side for output=⊖)

$\Longrightarrow$ purple line

$\Longrightarrow$ yellow line

$\Longrightarrow$ bright blue line

$\Longrightarrow$ dark blue line (built by finally adjusted weight) can divide the graph (one side for output=⊕, the other side for output ⊖)

$\Longrightarrow$ all predicted output P_y = output y (ground truth), and learning algorithm converge

D.

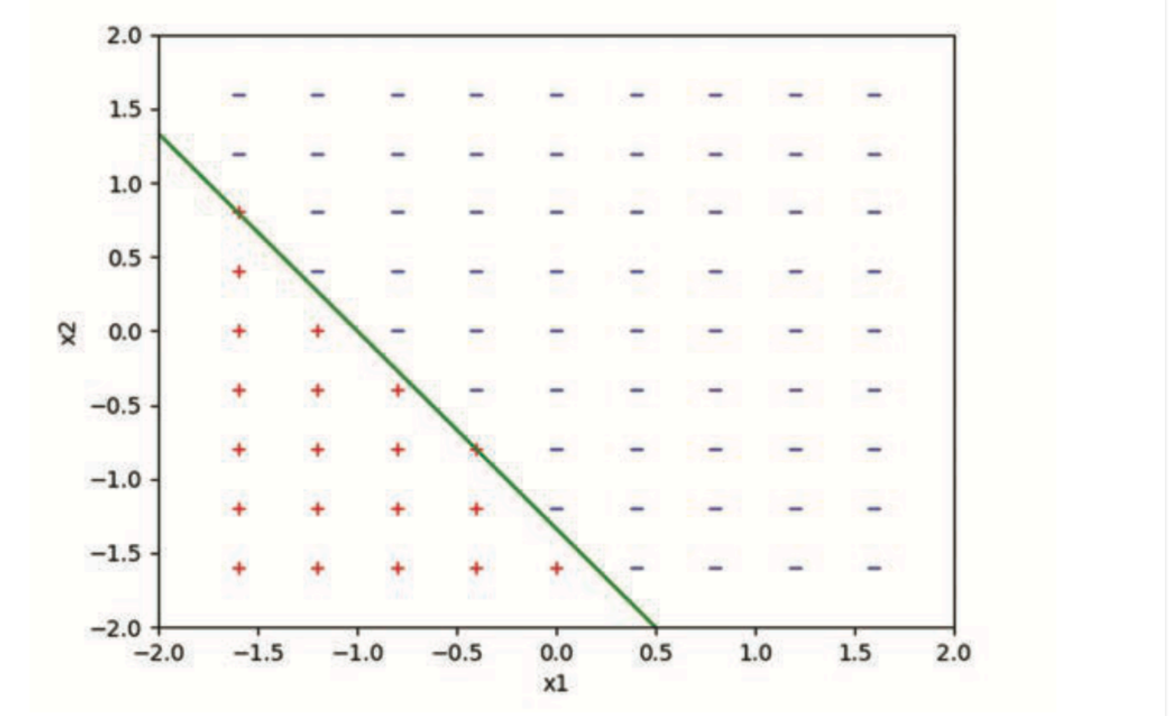


Figure 1-5 Output of a perceptron as a function of two inputs x_1 and x_2